

Seelam Avinash Reddy

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PROFESSIONAL SUMMARY

AI / Machine Learning Engineer (B.Tech, AI & ML) with demonstrated expertise in supervised learning, NLP, and computer vision through two production-ready internship projects achieving 90%+ accuracy. Built end-to-end ML pipelines from data preprocessing and feature engineering to model deployment. Proficient in MLOps workflows and Agentic AI concepts. Seeking AI/ML/Data Science roles to build scalable, production-ready models that drive measurable business outcomes.

EDUCATION

Bachelor of Technology (Computer Science & Engineering - AI & ML) | Aug 2021 - Jun 2025

Vellore Institute of Technology - CGPA: 7.12/10

TECHNICAL SKILLS

Python | scikit-learn | NLP (NLTK, TF-IDF) | Streamlit | Flask | Pandas | NumPy | Machine Learning | Computer Vision (OpenCV concepts) | Matplotlib/Seaborn | Git | SQL | Model deployment concepts | Agentic AI | MLOps concepts | Feature engineering | Model evaluation

EXPERIENCE

Machine Learning Intern - SmartInternz | Sep 1, 2025 - Oct 10, 2025

- Engineered 'Anemia Sense', a clinical ML diagnostic pipeline for anemia detection using 8 blood biomarkers (hemoglobin, RBC, MCV, MCH) from patient records.
- Built robust data preprocessing pipeline with missing-value imputation, feature normalization, and strategic train/test splits to prevent data leakage.
- Compared and optimized classification models (Logistic Regression, Random Forest, SVM) using grid search hyperparameter tuning; evaluated with accuracy, precision, recall, F1-score, and ROC/AUC achieving 91% accuracy with 0.93 AUC.
- Deployed Flask-based inference app (app.py) with serialized model (model.pkl) for real-time predictions; published complete codebase and artifacts on GitHub: github.com/Avinash3625/Anemia-Sense

Machine Learning Intern - EDUNET Foundation | Dec 2024 - Jan 2025

- Engineered an end-to-end SMS spam detection system using Kaggle dataset (5,000+ messages) with NLP preprocessing techniques (tokenization, lemmatization, stopword removal) and TF-IDF vectorization.
- Optimized SVM and Multinomial Naive Bayes classifiers through grid search hyperparameter tuning, achieving 90% accuracy with 92% precision and 90% recall on imbalanced datasets.
- Mitigated class imbalance using SMOTE oversampling technique, improving minority class F1-score by 15% and reducing false negatives by 20%.
- Deployed interactive Streamlit application enabling real-time spam classification with confidence scores; delivered comprehensive technical documentation for production handoff. Project GitHub: <https://github.com/Avinash3625/SMS-Spam-Detection>

PROJECTS

Finding Water Index Value using ML | Oct 2022

Developed multi-output regression model predicting water quality indices across 8 parameters using Random Forest and XGBoost, achieving R^2 score of 0.85. Implemented explainable AI framework (SHAP values) to identify top 5 contributing factors (pH, turbidity, dissolved oxygen), enabling stakeholders to understand model decisions. Built anomaly detection pipeline using Isolation Forest to flag contamination events 24 hours in advance, reducing response time by 40%.

Music Recommendation System | Jul 2023

Designed collaborative filtering system using matrix factorization (SVD) combined with content-based filtering on audio features (tempo, key, energy), improving recommendation relevance by 25% over baseline CF approach. Solved cold-start problem for new users by incorporating demographic signals and contextual metadata (listening time, mood tags, device type).

Farm Management using Drones (CV & ML) | Mar 2024

Prototyped drone-based crop monitoring solution using OpenCV for image segmentation and stress detection across 100+ aerial images covering 5 hectares. Applied color space analysis (HSV conversion) and threshold-based detection to identify crop health anomalies with 92% accuracy. Enabled targeted intervention strategies through heat-map visualization, potentially reducing water usage by 30% and fertilizer application by 25% through precision resource allocation.

SMS Spam Detection | Dec 2024

Built an SMS spam detection pipeline using Kaggle dataset with NLP preprocessing and TF-IDF feature engineering; trained and compared SVM (RBF kernel) and Multinomial Naive Bayes classifiers. Achieved 90% accuracy with 92% precision through hyperparameter tuning and SMOTE-based class balancing. Deployed Streamlit demo with real-time classification. GitHub: <https://github.com/Avinash3625/SMS-Spam-Detection>

Anemia Sense | Sep 2025

Designed a clinical ML pipeline for early anemia detection using 8 blood biomarkers; trained and compared Logistic Regression (L2 regularization), Random Forest (100 estimators), and SVM models achieving 91% accuracy with 0.93 AUC. Implemented comprehensive model evaluation with confusion matrix, precision-recall curves, and feature importance analysis. Built Flask-based demo inference app with model serialization (pickle) and published complete code, model artifacts, and documentation: github.com/Avinash3625/Anemia-Sense

CERTIFICATIONS

- Microsoft AI & ML Engineering Professional Certificate (Coursera) - Dec 2024
- Introduction to Artificial Intelligence (Coursera)

LANGUAGES & INTERESTS

Languages: English, Telugu | **Interests:** web technologies, Android updates, coding challenges